2015 F=ma Exam: Problem 10

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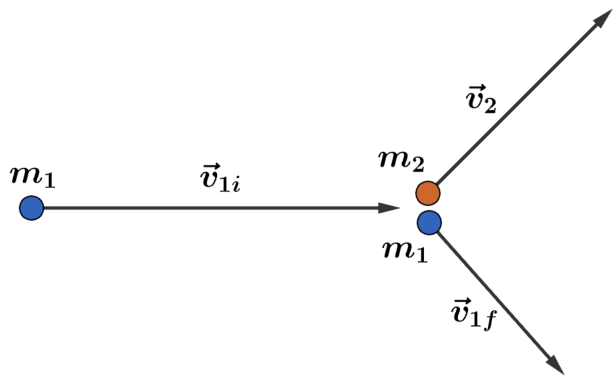

The initial kinetic energy is given by

$$K_i = \frac{1}{2}m_1 v_{1i}^2 = \frac{1}{2}(0.650\,\text{kg})(5\,\text{m/s})^2 = 8.13\,\text{J}$$

We found in the previous problem that $v_{1f} = 1.92\,\text{m/s}$. The final kinetic energy is then given by

$$K_f = \frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2 v_2^2 = \frac{1}{2}(0.650\,\text{kg})(1.92\,\text{m/s})^2 + \frac{1}{2}(0.750\,\text{kg})(4\,\text{m/s})^2 = 7.20\,\text{J}$$

and

$$\Delta K = K_f - K_i = -0.93\,\text{J}$$

Thus, we have

$$-K_i < \Delta K < 0$$

so the answer is $\boxed{\text{E}}$.